

Chemical equilibrium and routes to K

After midterm · Session 5 of 6 · 180 minutes

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Session outcome

Reconcile independent thermodynamic routes to K and then solve composition.

Before class: Review formation Gibbs energies, reaction enthalpy/entropy and integration of heat capacity.

Learning path and all labs

The 180-minute class

IN CLASS	MINUTES
Recall and prediction	10
Concepts and derivation	45
Worked example	30
Break	10
Instructor lab demonstration	25
Guided student exploration	35
Discussion and interpretation	15
Exit question and independent task	10

Reaction equilibrium changes composition

$$n_i = n_{i,0} + \nu_i \xi$$

$$\Delta_r G = \sum_i \nu_i \mu_i = \Delta_r G^\circ + RT \ln Q$$

Stoichiometric coefficients are negative for reactants and positive for products. At an interior equilibrium, $\Delta_r G = 0$.

K and Q have different roles

$$\ln K = -\frac{\Delta_r G^\circ}{RT}, \quad Q = \prod_i a_i^{\nu_i}$$

K is tied to T and the fixed reaction/standard-state convention. Q describes the supplied state.

Q=K is an equilibrium condition, not an identity for every measured composition.

Standard states keep K dimensionless

PHASE	ACTIVITY USED
Gas	$\phi_i y_i P/p^\circ$
Solute, molarity standard	$\gamma_i c_i/c^\circ$
Liquid solution	$\gamma_i x_i$
Present pure solid/liquid	1

Formation data and activities must use matching phases and standards.

Several routes to standard Gibbs energy

1. Supplied $\Delta_r G^\circ$ at the evaluation T.
2. Formation properties: $\Delta_r G^\circ = \sum \nu_i \Delta_f G_i^\circ$.
3. Reaction enthalpy and entropy: $\Delta_r G^\circ = \Delta_r H^\circ - T \Delta_r S^\circ$.
4. Reference K with integrated temperature dependence.
5. Equilibrium activities, only when equilibrium is supported.

Independent inputs can disagree; the lab preserves the disagreement.

A reference-state check

Lab 12 synthetic $A_2 \rightleftharpoons 2A$ at 350 K uses $\Delta_r G^\circ = -4.000$ kJ/mol.

$$\ln K = \frac{4000}{R(350)} = 1.37454, \quad K \approx 3.95326$$

With $\Delta_r H^\circ = 50.000$ kJ/mol and $\Delta_r S^\circ = 54000/350$ J mol⁻¹ K⁻¹, the H-TS route gives the same value.

Constant-enthalpy van 't Hoff relation

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\circ(T)}{RT^2}$$

With constant $\Delta_r H^\circ$ over the stated interval,

$$\ln \frac{K(T)}{K(T_r)} = \frac{\Delta_r H_r^\circ}{R} \left(\frac{1}{T_r} - \frac{1}{T} \right)$$

An endothermic reaction has increasing K with T in this approximation.

Constant heat-capacity correction

$$\Delta_r H^\circ(T) = \Delta_r H_r^\circ + \Delta_r C_p^\circ(T - T_r)$$

$$\Delta_r S^\circ(T) = \Delta_r S_r^\circ + \Delta_r C_p^\circ \ln(T/T_r)$$

The integrated $\ln K$ adds

$$\frac{\Delta_r C_p^\circ}{R} [\ln(T/T_r) + T_r/T - 1]$$

to the constant-enthalpy expression.

Reference consistency and temperature applicability

Changing only K_{ref} while holding reference H and S fixed creates two independent anchors that may disagree.

Point ΔG° and formation-energy data are unavailable away from their supplied T unless an explicit temperature model exists.

Keeping $\ln K$ avoids false zeros or infinities from exponent underflow/overflow.

Equilibrium extent and pressure

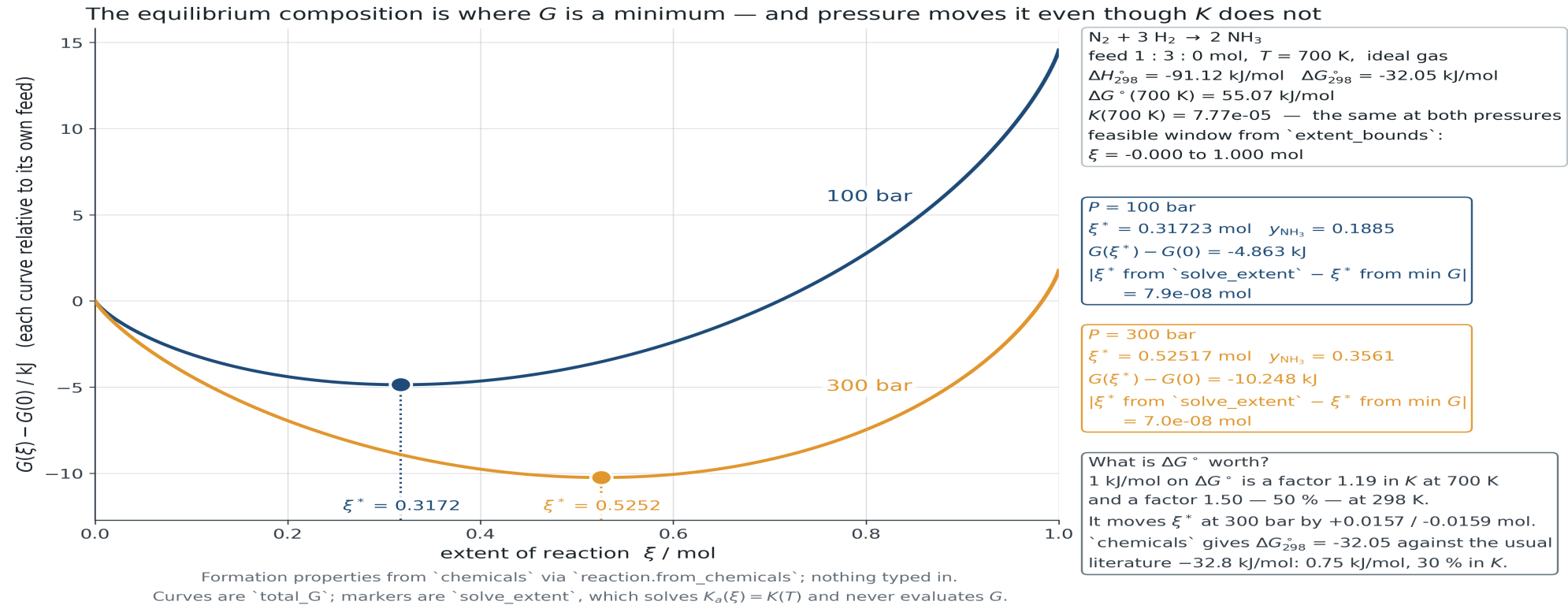
For a neutral ideal-gas reaction,

$$Q(\xi) = \prod_i [y_i(\xi)P/p^\circ]^{\nu_i}$$

Solve $\ln Q(\xi) - \ln K = 0$ within nonnegative mole amounts. Pressure changes Q and equilibrium composition, not standard K at fixed T .

Adding inert gas at fixed total pressure differs from adding it at fixed volume.

Gibbs energy along extent



Module 6 reference illustration: the admissible equilibrium is a constrained Gibbs-energy minimum.

Class demonstration • Lab 12

Open chemical equilibrium

1. Compare all applicable routes at the synthetic reference T.
2. Change T and compare ΔC_p and constant-property results.
3. Change pressure at fixed T; inspect extent and atom balances.
4. Export the study and restore it to repeat the calculation.

Guided exploration • 35 minutes

Choose one temperature change and one pressure change.

Keep a table of $\ln K$, Q , reaction direction and equilibrium extent. Explain which quantities change for each intervention.

Independently check one atom balance and one H-TS or van 't Hoff calculation.

Reaction scaling changes K

Reversing the reaction changes $\ln K$ to $-\ln K$. Multiplying every coefficient by c changes $\ln K$ to $c \ln K$.

Formation sums, reaction H , S , C_p and direct ΔG must transform on the same basis.

A numerical K without a written reaction is incomplete information.

Exit question

At the same temperature, pressure increases and equilibrium conversion changes.

Does this mean the equilibrium constant changed?

What additional information is needed before using one ΔG° value at a second temperature?

Independent practice • suggested 60–90 minutes

Compare constant-property and heat-capacity routes; report $\ln K$, Q and one extent/material-balance check.

Retain the calculator export, your worksheet, a comparison plot/table and one independent check. State an assumption that limits your conclusion.

Use the core labs on the [learning path](#). Optional extensions are additional work.

References and further study

Module reference deck · Lab sources and equations

IUPAC: standard electromotive force and standard equilibrium constant.

Synthetic worked examples illustrate calculations; they are not evidence of real-system accuracy.